Firmware & Software

This document summarizes what the shipped code does and how major pieces fit together. It is not a substitute for reading the sketches under `firmware/` or the Flutter sources under `android-app/vest_tracker/lib/`. For operator facing steps, see MobileApp.pdf and UserGuide.pdf. For pin maps and BOM, see Appendix.pdf.

FS1. Repository layout (reference)

Path	Role
`firmware/TRANSMITTER_VEST/`	Vest ESP32: GPS, rain analog, LoRa TX
`firmware/RECEIVER_CONTROL_TOWER_V2/`	Receiver ESP32: LoRa RX, Wi-Fi, Firebase RTDB upload, message queue, Malaysia `time_my`
`firmware/RECEIVER_CONTROL_TOWER/`	Earlier receiver variant (simpler, prefer V2 for production)
`firmware/*_testing_only/`	Bring-up sketches (GPS, LoRa, rain, Firebase isolation)
`android-app/vest_tracker/`	Flutter Vest Tracker Android app
`hardware/schematic/`	KiCad schematic (`vest_tracker_schematic.kicad_sch`)

Security note: receiver `.ino` files contain placeholder Wi-Fi and Firebase constants. Replace them with deployment-specific values before flashing. Do not commit real production secrets to a public repository.

FS2. Vest unit (transmitter)

Source: `firmware/TRANSMITTER_VEST/TRANSMITTER_VEST.ino`

Responsibilities

1. Parse NEO-6M GPS NMEA on ESP32 UART2 (RX `GPIO16`, TX `GPIO17`, 9600 baud).
2. Sample rain sensor analog on GPIO34 with debouncing.
3. Transmit compact JSON over LoRa on a timer and on wet edges.

4. Emit human-readable USB Serial JSON (10 s) for maintenance 115200 baud.

Compile-time identity: `VEST_ID` (default `1`) > maps to RTDB `device_01` at receiver.

GPS stage state machine

`gs`	Constant	`gsn` string	Entry condition
`1`	`GPS_STAGE_WAITING_FOR_SIGNAL`	`waiting_for_signal`	No GPS bytes within 2000 ms (`GPS_NO_DATA_TIMEOUT_MS`)
`2`	`GPS_STAGE_AVAILABLE_NO_LOCK`	`no_satellite_lock`	Bytes present but no valid fix, or fix age ≥ 2000 ms
`3`	`GPS_STAGE_LOCKED`	`locked`	Valid lat/lon and fix age < 2000 ms

`lastGoodFix` behavior: When stage 3 is reached, firmware stores lat, lon, alt, speed, satellites, fix timestamp. LoRa payloads prefer this stored snapshot even if live stage later drops to 1 or 2, operators may see stale coordinates with fresh wet status until a new outdoor lock.

LoRa transmit timing

Trigger	Interval / rule
Scheduled TX	Every 60 s (`LORA_TX_INTERVAL_MS`)
Event TX	Immediately on debounced dry to wet edge
Rain sample skip	Skipped on 60 s uptime boundary and when scheduled LoRa TX is due (avoids ADC read during RF)
Serial JSON skip	Also suppressed on 60 s boundary tick

No deep sleep, vest runs continuously with 1 ms loop delay. Boot delay 1000 ms. LoRa init failure to infinite hang (“LoRa INIT FAILED”).

Rain detection

Constant	Value	Meaning
`RAIN_ANALOG_WET_THRESHOLD`	`2500`	Wet when `analogRead(GPIO34) < 2500`
`RAIN_DEBOUNCE_MS`	`80`	Stable state must hold 80 ms before change

1. `wetEdgeToTrue`: set only on transition dry to wet (used to trigger immediate LoRa TX).
2. `imp` (important): in current logic equals stable wet state (true whenever wet, not edge-only).

JSON formats

1. USB Serial (diagnostics, 10 s, buffer 512 bytes)
    a. Long field names: `gps_stage`, `gps_stage_name`, `latitude`, `longitude`, `altitude_m`, `speed_kmh`, `satellites`, `fix_age_ms`, `water_detection`, `important_event`, `uptime_ms`.
    b. Lat/lon precision: 6 decimal places.

2. LoRa (compact radio, buffer 240 bytes)
    a. With stored fix: {"id":1,"gs":3,"gsn":"locked","lat":14.59950,"lon":120.98420,"alt":15,"spd":0.0,"sat":7,"fix_ms":123456,"wet":false,"imp":false,"t":654321}
    b. Without stored fix: {"id":1,"gs":1,"gsn":"waiting_for_signal","loc":false,"wet":false,"imp":false,"t":654321}

Field	Meaning
`id`	Vest numeric ID to RTDB path
`gs`, `gsn`	GPS stage number and name
`lat`, `lon`, `alt`, `spd`, `sat`, `fix_ms`	Present when including position snapshot

Field	Meaning
`loc`	When `false`, receiver treats location as absent
`wet`	Rain boolean
`imp`	Important flag (stable wet in current logic)
`t`	Transmitter uptime (ms)

LoRa radio: 433 MHz (`433E6`), SPI pins SS=`5`, RST=`14`, DIO0=`2`, SCK/MISO/MOSI =`18`/`19`/`23`.

FS3. Control tower (receiver V2)

Source: `firmware/RECEIVER_CONTROL_TOWER_V2/RECEIVER_CONTROL_TOWER_V2.ino`

Responsibilities

1. LoRa RX on same frequency/pinout as vest.
2. Parse LoRa JSON (`parseVestJson`, `StaticJsonDocument<512>`).
3. Enqueue messages in RAM, flush to Firebase when Wi-Fi and Firebase ready.
4. NTP sync to attach `time_my` (Malaysia UTC+8, ISO-8601 `YYYY-MM-DDTHH:MM:SS+08:00`).
5. Resilience: exponential backoff, token-not-ready handling, hard recovery after repeated RTDB failures.

Message queue

Constant	Value
`QUEUE_CAPACITY`	30 messages
`MAX_FLUSH_PER_LOOP`	4 RTDB writes per loop iteration

Ring buffer: when full, oldest message is dropped before push. LoRa RX continues even when Wi-Fi is down, extended outages can lose historical snapshots (RTDB always stores latest state per device, not full history).

Wi-Fi / Firebase / RTDB resilience

Three `Backoff` instances (`wifiBackoff`, `firebaseBackoff`, `rtdbWriteBackoff`):

Initial delay 250 ms, max 30000 ms, exponential doubling.

Other timing:

Constant	Value	Purpose
Wi-Fi re-`begin` interval	60000 ms	If disconnected too long
NTP timeout	20000 ms	`pool.ntp.org`, `time.nist.gov`, UTC+8
Firebase socket/response timeout	20000 ms each	
Firebase Wi-Fi reconnect timeout	10000 ms	
`POST_FIREBASE_BEGIN_QUIET_MS`	7000 ms	No RTDB writes immediately after `Firebase.begin()`
Token-not-ready wait	5000 ms	Does not count as hard RTDB failure
`MAX_CONSECUTIVE_RTDB_FAILURES_BEFORE_HARD_RECOVERY`	6	Triggers full recovery
`HARD_RECOVERY_COOLDOWN_MS`	60000 ms	Between hard recoveries

1. Hard recovery: reset Firebase session, disconnect Wi-Fi, clear NTP synced flag, reset all backoffs, clears stuck SSL/TCP state.
2. RTDB write API: single atomic `Firebase.RTDB.updateNode` per message (not per-field sets).

LoRa > RTDB field mapping

Path rule: `/vest/device_XX` where `XX` is zero-padded to two digits when `id < 10` (`id=1` is `device_01`, `id=10` is `device_10`).

LoRa / source	RTDB key	Notes
`id`	path segment	See rule above
`gs`	`gs`	default 0 if missing
`gsn`	`gsn`	max 31 chars
`wet`	`wet`	bool
`imp`	`imp`	bool
`t`	`t`	uint32 uptime ms
`lat`, `lon`, `alt`, `spd`, `sat`, `fix_ms`	same keys	Only if lat+lon present and `loc` not explicitly `false`
receiver NTP	`time_my`	Omitted if NTP never synced

Example RTDB documents (after V2 upload)

Locked with location:

{
  "gs": 3,
  "gsn": "locked",
  "lat": 14.5995,
  "lon": 120.9842,
  "alt": 15,
  "spd": 0.0,
  "sat": 7,
  "fix_ms": 123456,

"wet": false,
"imp": false,
"t": 654321,
"time_my": "2026-05-30T14:32:01+08:00"
}

No fix:

{
"gs": 1,
"gsn": "waiting_for_signal",
"wet": false,
"imp": false,
"t": 654321,
"time_my": "2026-05-30T14:32:01+08:00"
}

Optional `deviceName` may be added manually in Firebase for prettier app labels.

FS4. Receiver V1 vs V2 (prefer V2)

Area	V1 (`RECEIVER_CONTROL_TOWER`)	V2 (`RECEIVER_CONTROL_TOWER_V2`)
`time_my` in RTDB	NTP synced but not written	Written on every upload
RTDB write	Multiple `setInt`/`setString`/… calls	Single `updateNode` JSON
RTDB backoff	None	Dedicated exponential backoff
Hard recovery	None	After 6 consecutive failures

Area	V1 (`RECEIVER_CONTROL_TOWER`)	V2 (`RECEIVER_CONTROL_TOWER_V2`)
Token-not-ready	Counted as failure	Detected, 5 s wait, no failure increment
Post-`Firebase.begin()` delay	None	7 s quiet period
Wi-Fi sleep	Auto-reconnect only	Also `WiFi.setSleep(false)`

Both share: queue capacity 30, LoRa 433 MHz, same SPI map, flush up to 4/loop.

FS5. Mobile application (Flutter)

Source: `android-app/vest_tracker/lib/`

1. Architecture (summary)
    a. No Provider/Riverpod: config via `AppConfigScope` (`SharedPreferences`), Firebase via `FirebaseClientScope`.
    b. `VestRepository`: streams `vest/` and parses device models (`vest_models.dart`).
    c. Auth gate: signed-in `auth.uid` must equal `allowedUid` in app config.
    d. Map: `flutter_map` + OpenStreetMap tiles; markers from RTDB lat/lon.
    e. Notifications: `flutter_local_notifications`, channel `wet_alerts`.
    f. Background: WorkManager task `vest_tracker_wet_monitor`, frequency from `pollIntervalMinutes` (clamped 15–240).

2. Wet alert logic (app layer)
    a. Foreground: `WetMonitorForeground` RTDB stream while app open.
    b. Background: `WetMonitorBackground` periodic poll.
    c. Notify only on `wet` false → true; first read per device sets baseline in prefs (`vest_tracker.lastWet.{deviceId}`).
    d. Scope: all devices if `selectedDeviceId` empty, else single device.

Online heuristics (support note)

Screen	Window	Use
Live Map marker ripple	5 minutes since last seen	“Online” animation
Devices card label	10 minutes	Online/Offline text

Last seen prefers `time_my` (Malaysia ISO) when present, otherwise falls back to other timestamp fields in model parsing.

Optional log upload

When `logsUploadEnabled`, `AppLogger` may push entries to RTDB path `appLogs/{clientInstanceId}/…`.

For screen-by-screen UI detail, (see MobileApp.pdf).

FS6. Normal runtime expectations

Event	Typical timing
Scheduled LoRa TX	~60 s per vest
Immediate LoRa TX	Debounced dry to wet
USB Serial JSON (vest)	~10 s (when USB connected)
GPS cold start	Few minutes outdoors
Cloud visibility	Receiver on + Wi-Fi + valid Firebase credentials
Background wet poll (app)	15–240 min (user setting; OS may throttle)

FS7. Reset behavior

1. Vest power-cycle: GPS acquisition restarts. `lastGoodFix` cleared on boot, first LoRa may omit coordinates until new lock.
2. Receiver power-cycle: Uploads pause until Wi-Fi, NTP (for `time_my`), and Firebase ready; queued messages may flush afterward (subject to 30-message capacity).

FS8. Constants quick reference for transmitter

Constant	Value
`VEST_ID`	`1` (per deployment flash)
`GPS_JSON_INTERVAL_MS`	`10000`
`LORA_TX_INTERVAL_MS`	`60000`
`GPS_NO_DATA_TIMEOUT_MS`	`2000`
Fix lock threshold	fix age < 2000 ms
`RAIN_DEBOUNCE_MS`	`80`
`RAIN_ANALOG_WET_THRESHOLD`	`2500`
`LORA_FREQ_HZ`	`433E6`
Serial / LoRa JSON buffers	`512` / `240` bytes

FS9. Constants quick reference for receiver V2

Constant	Value
`QUEUE_CAPACITY`	`30`
`MAX_FLUSH_PER_LOOP`	`4`

Constant	Value
Backoff initial / max	`250` / `30000` ms
NTP timeout	`20000` ms
`POST_FIREBASE_BEGIN_QUIET_MS`	`7000` ms
Token-not-ready wait	`5000` ms
Max RTDB failures before hard recovery	`6`
Hard recovery cooldown	`60000` ms
`gsn` max length	`31` chars
`time_my` buffer	`32` chars

FS10. Firmware updates (maintenance)

Firmware is updated by maintainers using Arduino IDE (or compatible tooling), the ESP32 board support package, and library versions referenced by each sketch. Operators should not reflash unless instructed. When reporting issues, include: device ID, symptom, approximate time, and any Serial Monitor capture at 115200 baud.

Data flow overview: (see SystemArchitecture.pdf). Fault finding: (see TroubleshootingGuide. pdf).